Introduction to Embodied AI

Lecture 2: Embodied AI System Overview

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Agenda

- The Embodied AI paradigm
- Embodied AI System Design
- Sensors Overview
- Robot Vision vs. Computer Vision
- Robot Demos from my lab

What is Embodied AI?

- Embodied AI is the intersection of **machine learning and robotics**
- The study of **training optimal solutions using sensory data** for robots to act and achieve goals
- Perception and decision making are the key
- Possible future
    - General-purpose home robots!

The Embodied AI Paradigm

- Let's start with - How do we make a self-driving vehicle?
  - Given
    - Starting location, Goal location
    - Map of the city
    - Sensors - Camera, LIDAR, etc.
  - Objective
    - Minimize distance to goal
  - Constraint
    - Follow the road
    - Follow traffic rules
    - No crashing into other objects

The Embodied AI Paradigm

- How do we make a self-driving vehicle? - here is a flow chart
    - Actions includes steering wheel and pedals

?

Sensors

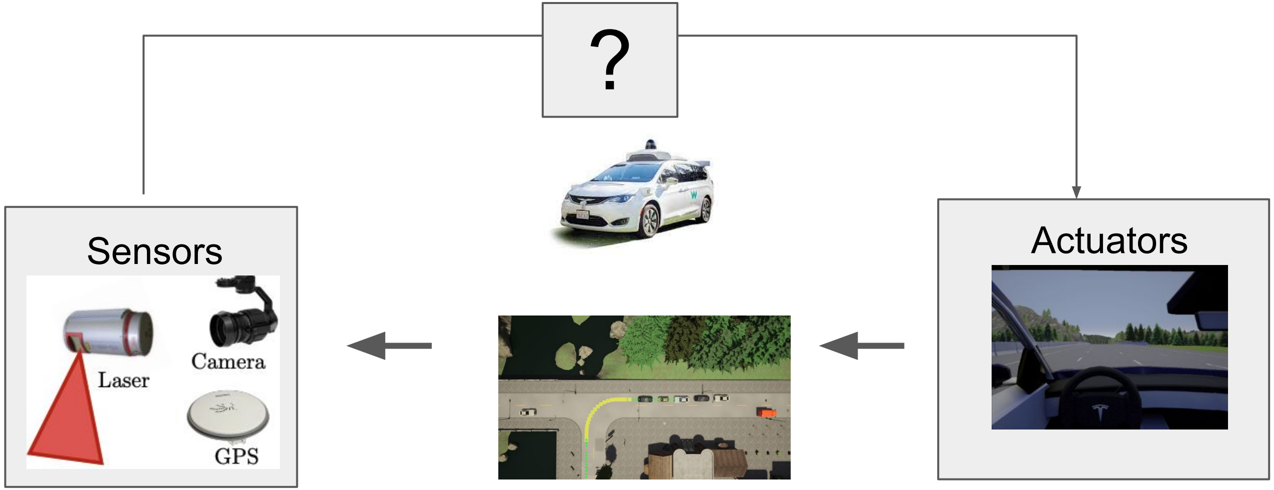
Robot interacts
with environment

Actuators

Steering wheel, pedals

The Embodied AI Paradigm

- Key questions
  - Q1: **Assume we know everything about the world.** What commands should we send to the actuator?
  - Q2: How do we update the world knowledge given the sensors?



The Embodied AI Paradigm

- Suppose we have a **world model**

Note in robotics, people use “model” to refer to a representation of the physical system. In ML, the model refers to an algorithm (and its parameters).

World Model

?

?

Sensors

Robot interacts with environment

Actuators

Steering wheel, pedals

The Embodied AI Paradigm

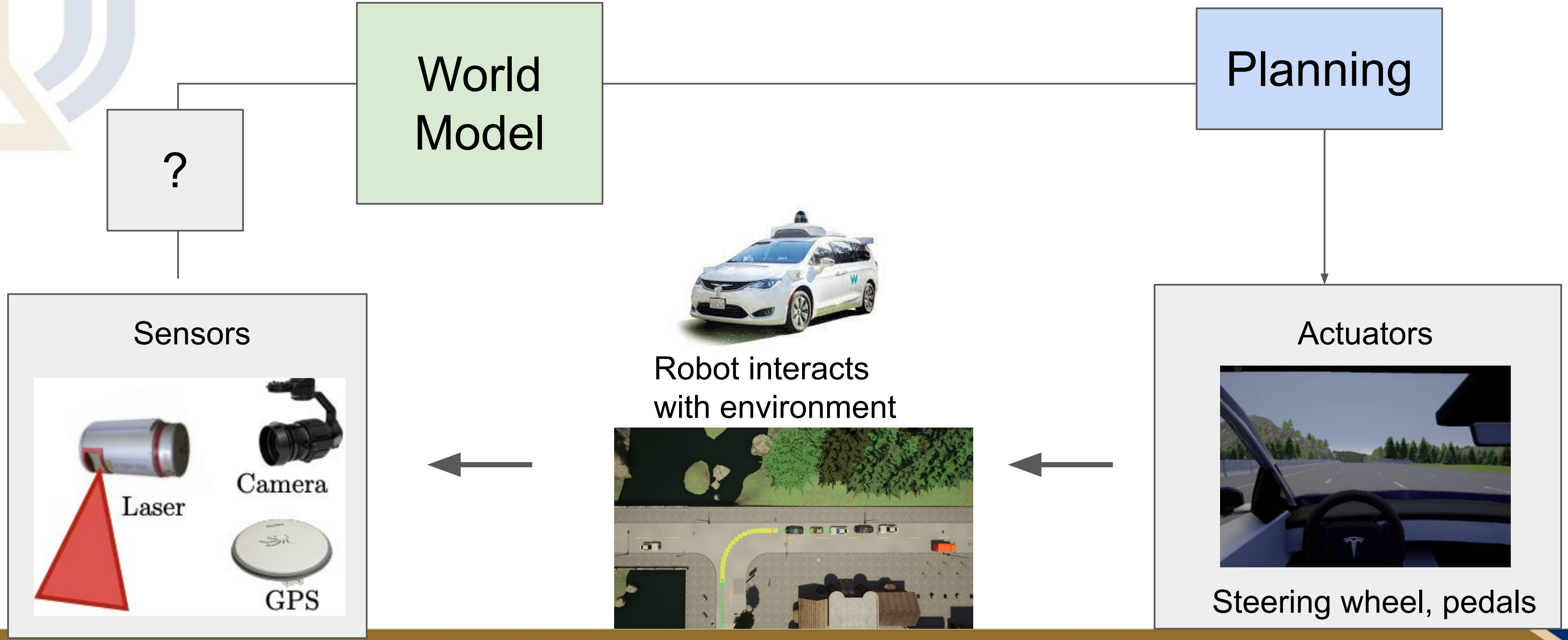
- What is the world model? It includes:
  - **Where is the robot/vehicle? What is its current status?**
  - What are obstacles (sidewalk, grass, trees) and what are available to drive (road)?
  - What objects to avoid (other vehicle, pedestrians)?
  - What are the properties of the vehicle? (Car Model)
    - How fast it can go? Acceleration rate?
    - Wet ground?
  - Nowadays, world model means "pixel and action prediction" model. So the vision generative models are considered world model

Note in robotics, people use "model" to refer to a representation of the physical system. In ML, the model refers to an algorithm (and its parameters).

In essence, world model means a "state transition" model $P(s_t, a_t)=s_{t+1}$
It predicts the future "state"

The Embodied AI Paradigm

- Given the world knowledge, we can then do planning!



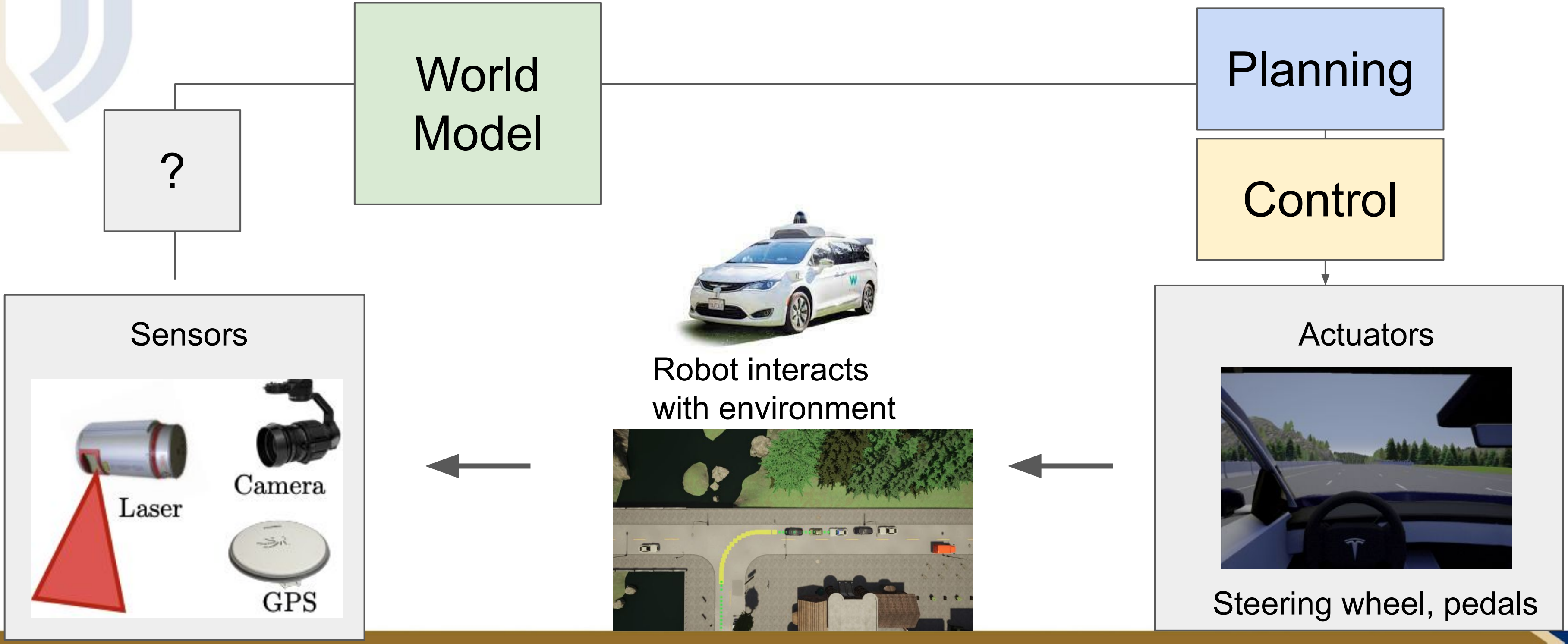
The Embodied AI Paradigm

- What is planning?
  - **Planning is an optimization problem in which we find the optimal sequence of actions towards minimizing a cost function**
    - In the self-driving case, minimizing the robot’s distance to the goal location
  - While satisfying constraints (no crashing, follow traffic rules)
  - There are global planning (global route) and local planning (change lane? Turn?)

The Embodied AI Paradigm

- Given the plan, what about the control?

The plan gives where to go. Now we need t
control the robot to do so



The Embodied AI Paradigm

- What is planning and **control**?
  - There are global (global route) and local planning (change lane? Turn?)
  - **A controller will execute the plan by giving specific commands to the actuators**
    - For example, convert the trajectory/motion plan (which way the car should go and how fast) into the **steering** and **pedal** command

The Embodied AI Paradigm

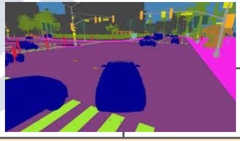
- Key questions
  - Q1: Assume we know everything about the world. What commands should we send to the actuator?
    - Done. We know how to control the car if we know about the world
  - **Q2: How do we update the world knowledge given the sensors?**

The Embodied AI Paradigm

- **State Estimation -** Update the world knowledge given the sensors
  - Given the sensor data, the perception module
    - extracts necessary information (about the surroundings)
    - localizes the robot in the world

The Embodied AI Paradigm

- How do we make a self-driving vehicle? - Here is the (simplified) completed graph



State Estimation

World Model

Planning

Control

Sensors

Robot interacts with environment

Actuators

Steering wheel, pedals

CMU’s winning entry to DARPA Self-Driving challenge 2007

- Let’s look at a successful self-driving system from researchers at CMU

https://www.youtube.com/watch?v=3dD9HVNIIG0

Tartan Racing: A Multi-Modal Approach to the DARPA Urban Challenge

April 13, 2007

Chris Urmson, Joshua Anhalt, Drew Bagnell, Christopher Baker, Robert Bittner,
John Dolan, Dave Duggins, Dave Ferguson, Tugrul Galatali, Chris Geyer, Michele Gittleman,
Sam Harbaugh, Martial Hebert, Tom Howard, Alonzo Kelly, David Kohanbash, Maxim Likhachev,
Nick Miller, Kevin Peterson, Raj Rajkumar, Paul Rybski, Bryan Salesky, Sebastian Scherer,
Young Woo-Seo, Reid Simmons, Sanjiv Singh, Jarrod Snider, Anthony Stentz,
William "Red" Whittaker, and Jason Ziglar

Carnegie Mellon University

Hong Bae, Bakhtiar Litkouhi, Jim Nickolaou, Varsha Sadekar, and Shuqing Zeng

General Motors

Joshua Struble and Michael Taylor

Caterpillar

Michael Darms

Continental AG

*One of the most well-documented autonomous driving system

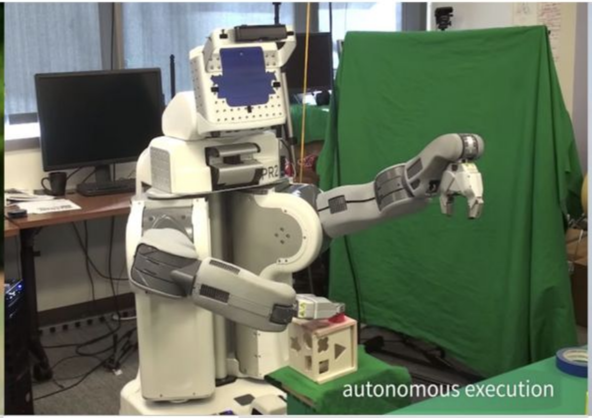
CMU’s winning entry to DARPA Self-Driving challenge 2007

- See the number of sensors on top of the vehicle
- And the 3D visualization of mapping and localization of the car

The Embodied AI Paradigm

- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
  - Drones, robotic arms, legged machines

[Sa et al. IROS 2014]

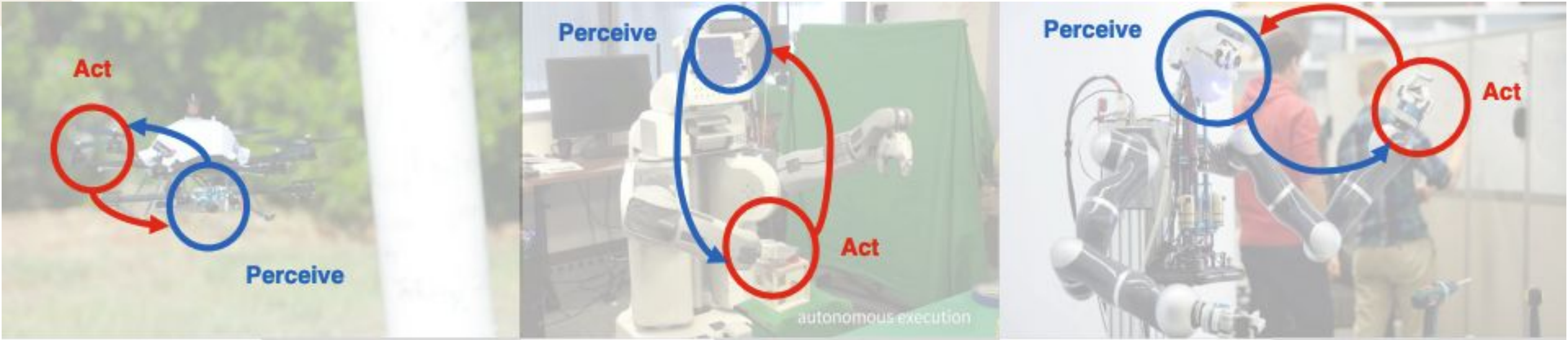


[Levine et al. JMLR 2016]

[Bohg et al. ICRA 2018]

The Embodied AI Paradigm

- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
  - The key is the perception-action loop

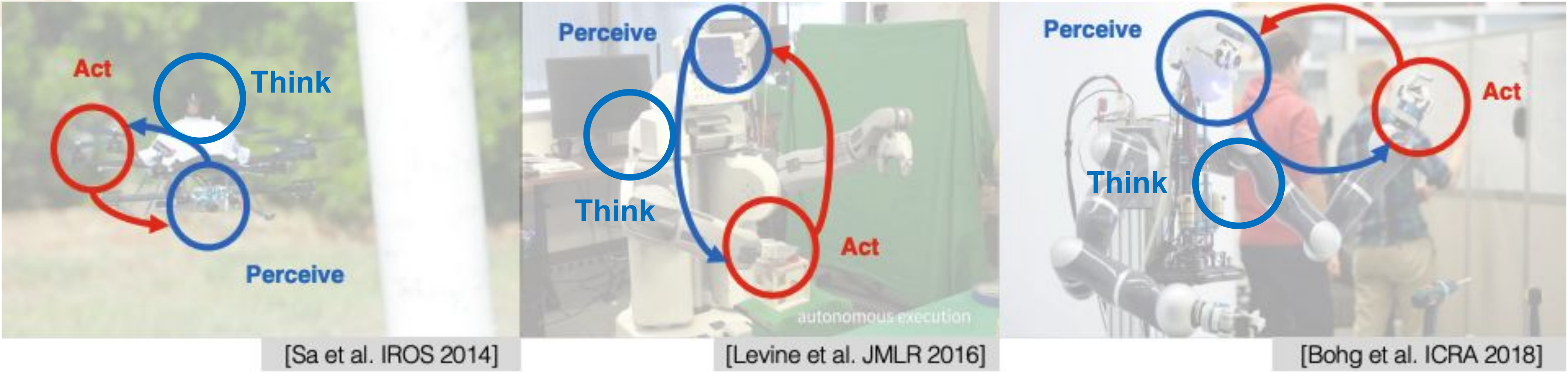


[Sa et al. IROS 2014] [Levine et al. JMLR 2016] [Bohg et al. ICRA 2018]

Perceive - Sensors are similar: Cameras, LIDARs, Inertial Measurement Units (IMUs)
Act - Actuators: Joints, arms, claws, motors

The Embodied AI Paradigm

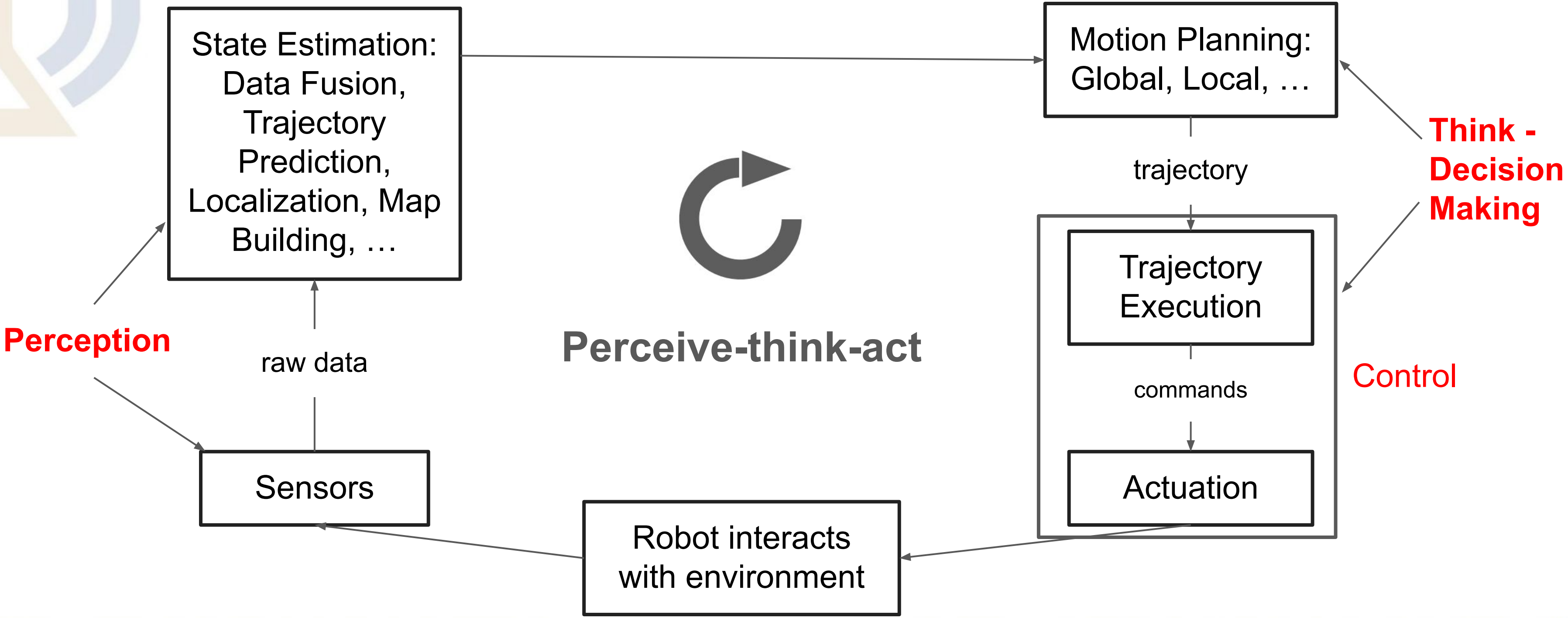
- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
  - Adding **machine learning** to close the perception-action loop (we call the process "Think")



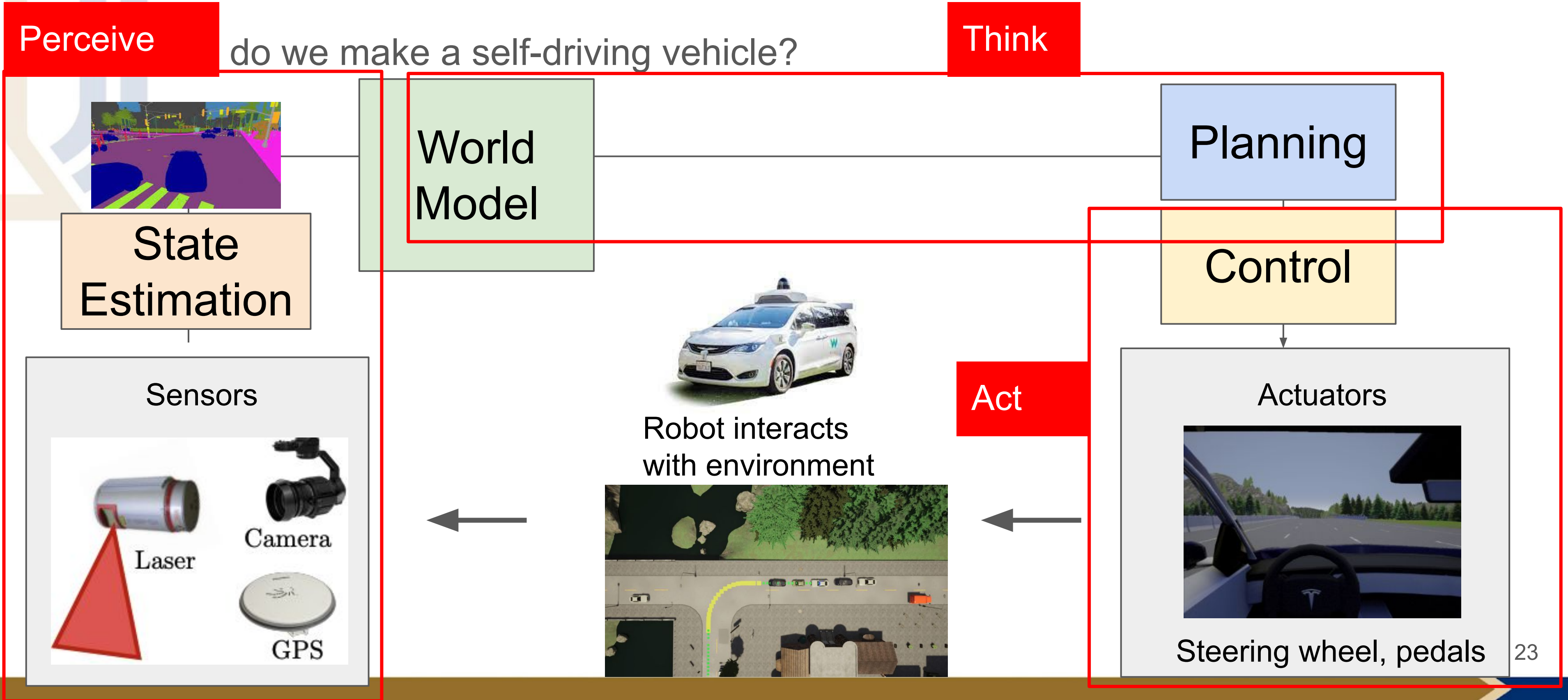
Perceive - Sensors are similar: Cameras, LIDARs, Inertial Measurement Units (IMUs)
Act - Actuators: Joints, arms, claws, motors

The Embodied AI Paradigm

- A more generalized paradigm - The Perceive-think-act circle



The Perceive-Think-Act Cycle in previous self-driving system

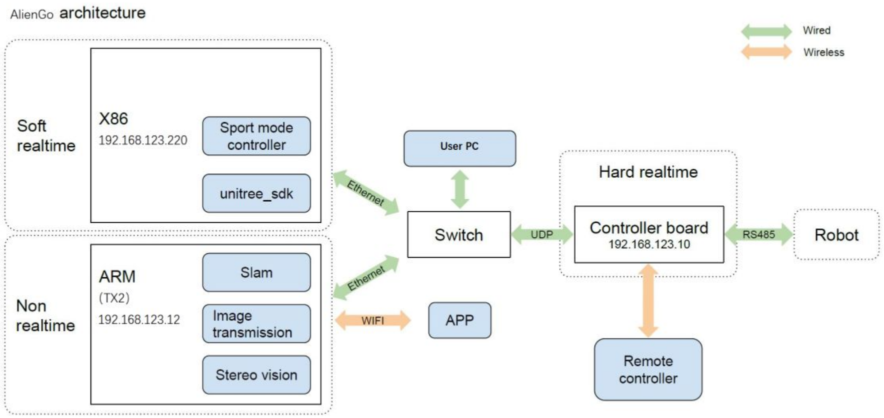


Agenda

- The Embodied AI paradigm
- **Embodied System Design**
- Sensors Overview
- Robot Vision vs. Computer Vision

Embodied System Design

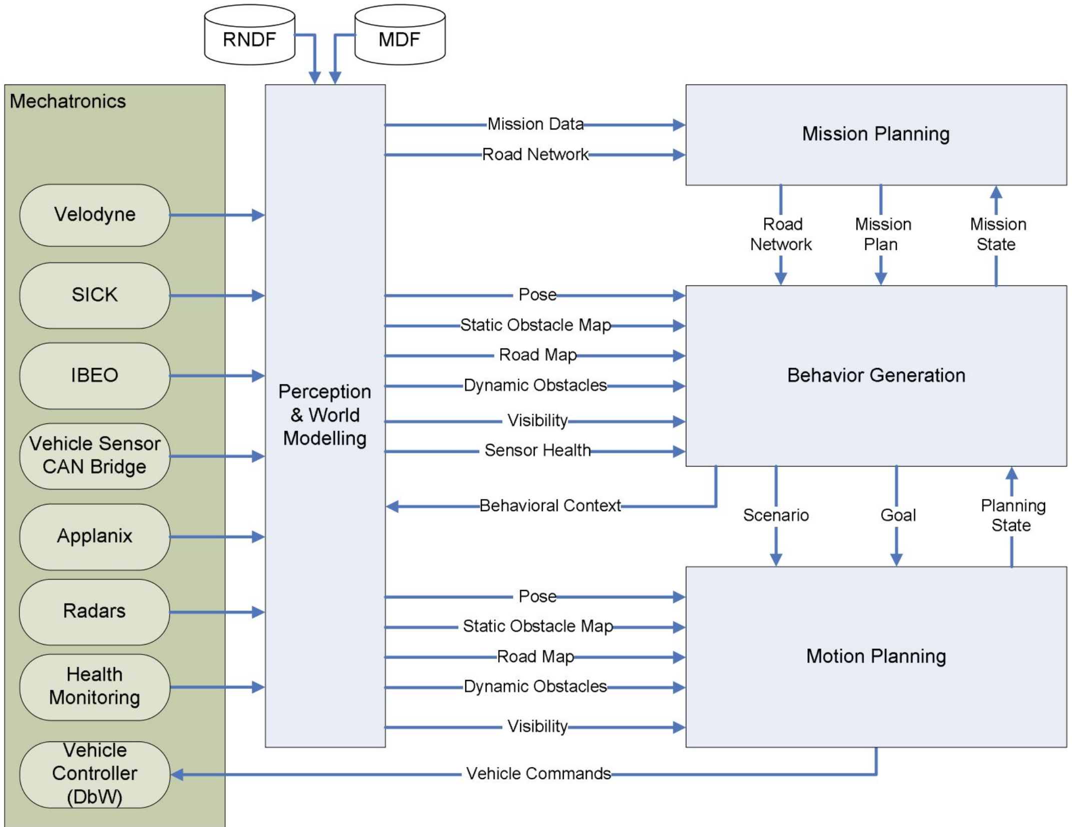
- The hardware
  - Sensors
  - Actuators
- The software
  - Lowlevel: Control Unit Drivers
  - Middleware: Robot Operating System (ROS)
    - Communication between modules
    - Data logging
    - Visualization
  - Perception models
  - Motion planning models
- Principles
  - Follow the Perceive-think-act paradigm
  - Modular, robust, data logging and playback



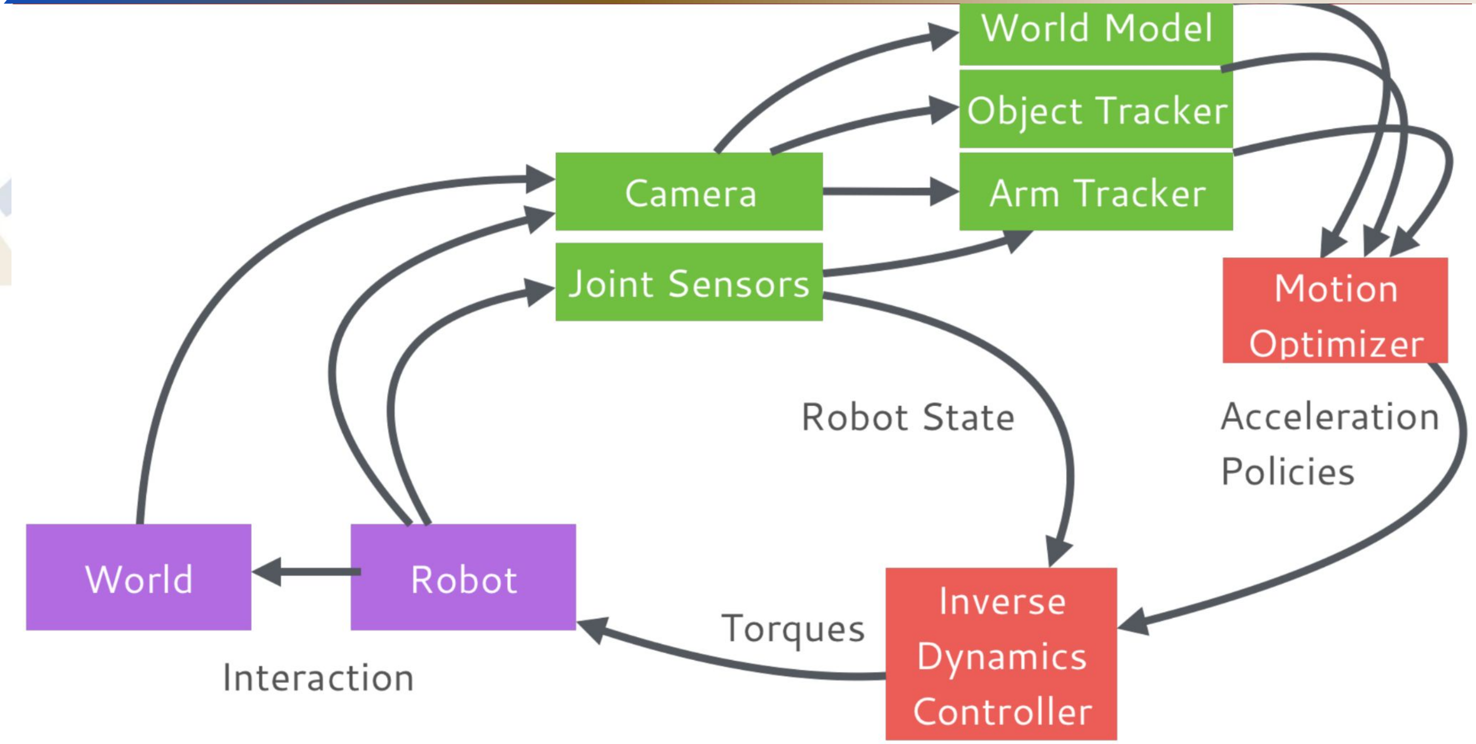
The architecture of Unitree's four legged-robot

Embodied System Design

- Example: CMU's Self-driving system
  - Perception
  - Planning
  - Mechatronics (Controller & sensors)



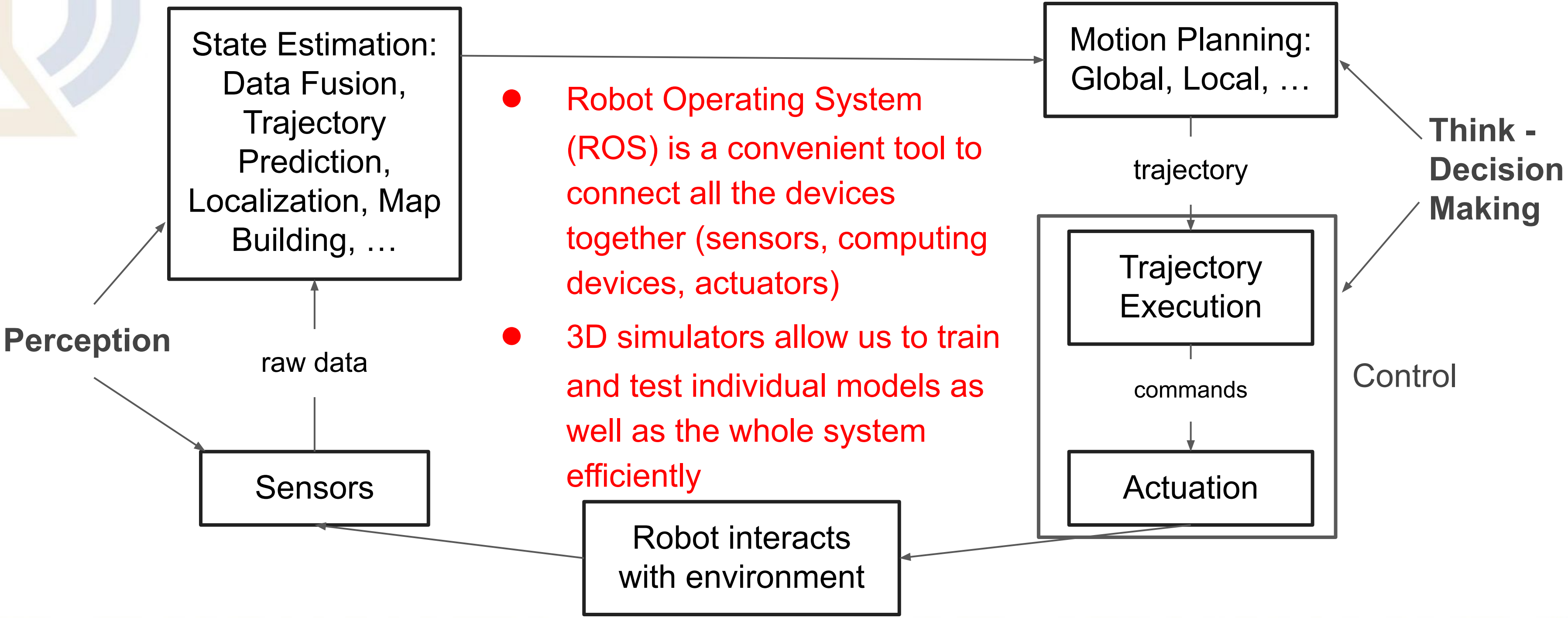
Embodied System Design



- Example: Kappler et. al. Real-Time Perception Meets Reactive Motion Generation. ICRA 2018

Embodied System and The Embodied AI Paradigm

- ROS and 3D Simulator are key component for Embodied AI



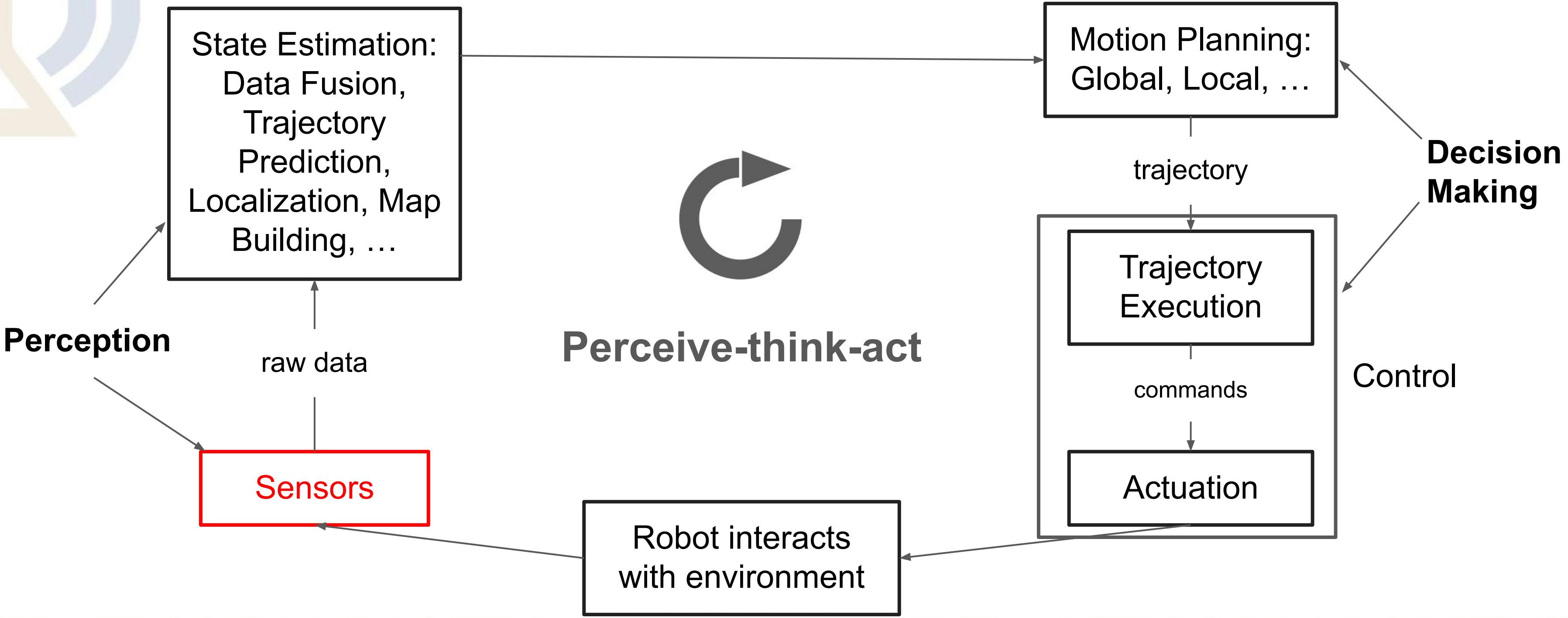
- Robot Operating System (ROS) is a convenient tool to connect all the devices together (sensors, computing devices, actuators)
- 3D simulators allow us to train and test individual models as well as the whole system efficiently

Agenda

- The Embodied AI paradigm
- Embodied System Design
- **Sensors Overview**
- Robot Vision vs. Computer Vision

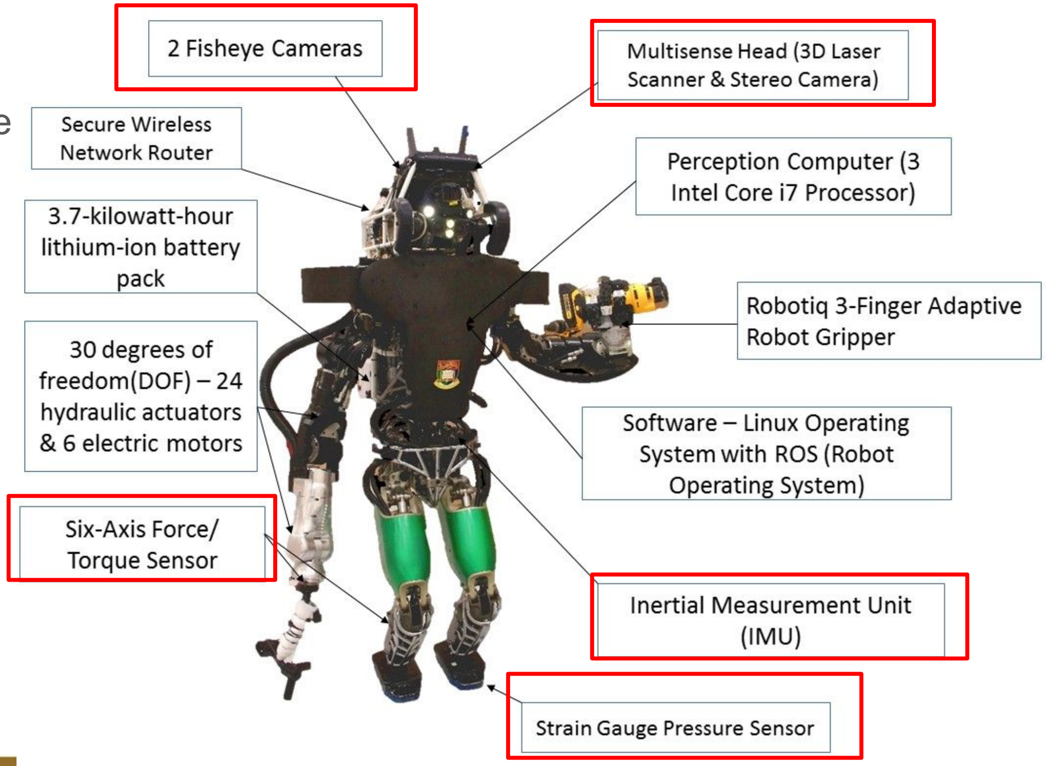
The Embodied AI Paradigm

- The Perceive-think-act circle



Robotic Sensors

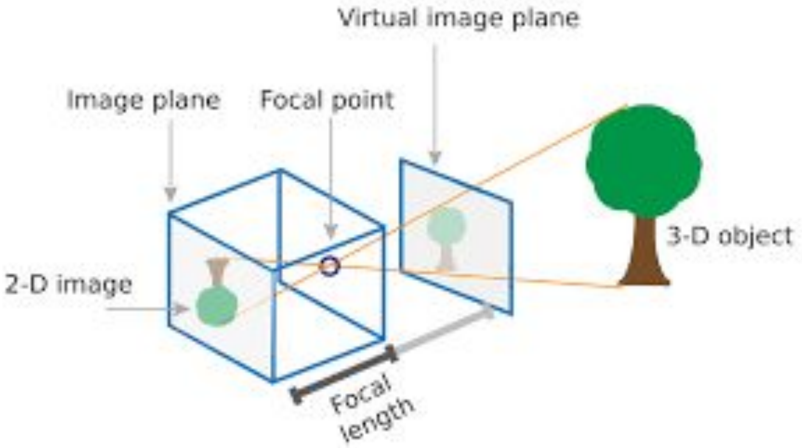
- Multimodal sensors are key to robot perception
  - Example: DARPA robotic challenge 2015
    - Fisheye cameras
    - LIDAR
    - Inertial Measurement Unit (IMU)
    - Torque Sensor
    - Pressure Sensor



[Source: HKU Advanced Robotics Laboratory]

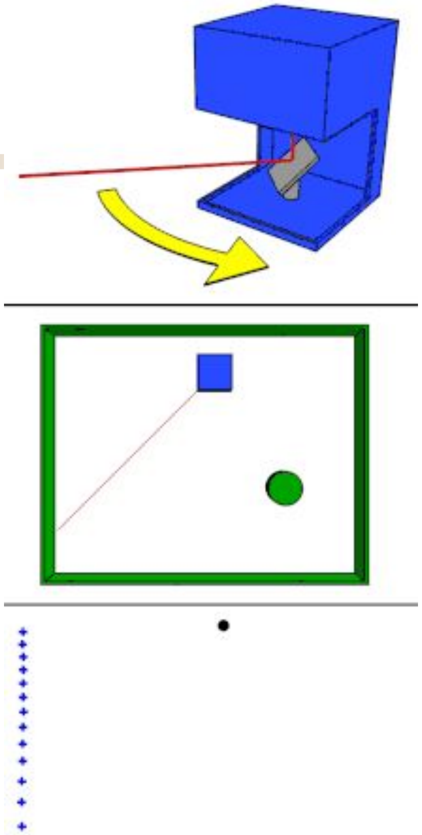
Robotic Sensors

- Multimodal sensors are key to robot perception
  - Cameras, Stereo Cameras
    - How does the 3D world project to a 2D image?
    - We will learn about it in lecture 4!

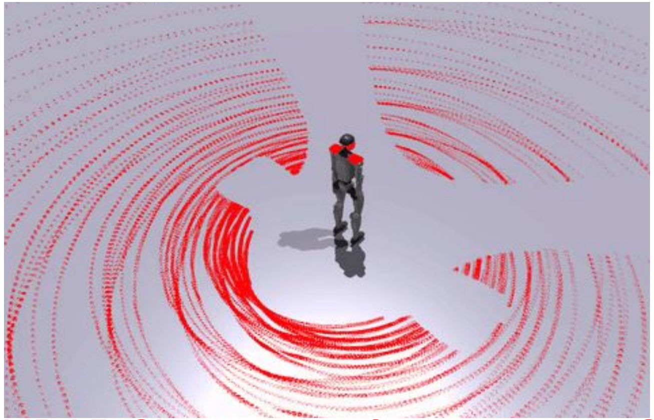


Robotic Sensors

- Multimodal sensors are key to robot perception
  - Lidar
    - Short for light detection and ranging
    - Send laser and get distance to obstacles
    - Why the discontinuance?
      - You can only sample finite angles



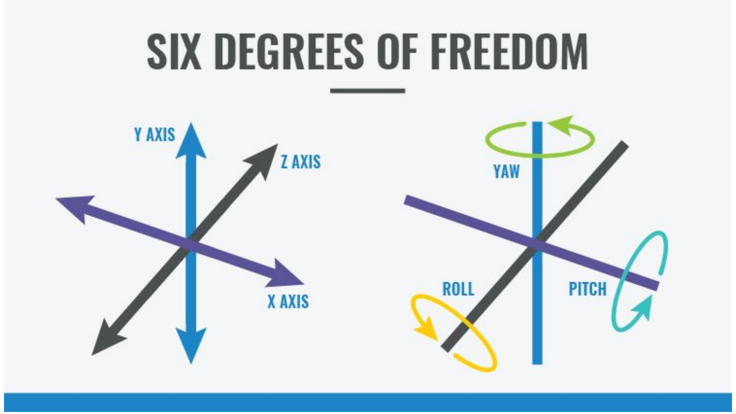
LiDAR Sim from CARLA

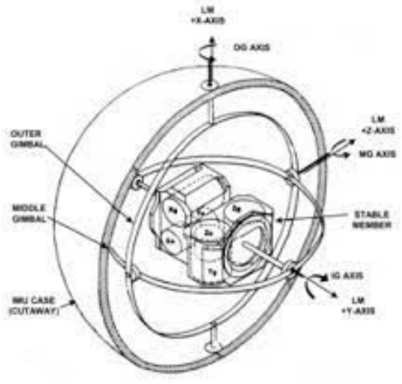


LiDAR Sim on robot from our work: OmniPerception-CoRL’25

Robotic Sensors

- Multimodal sensors are key to robot perception
  - Inertial Measurement Unit (IMU)
    - Measure angular rate, orientation, acceleration, etc.
    - Used in drones, and smartphones!

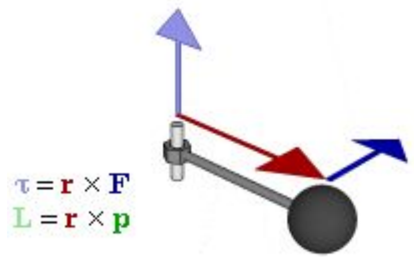


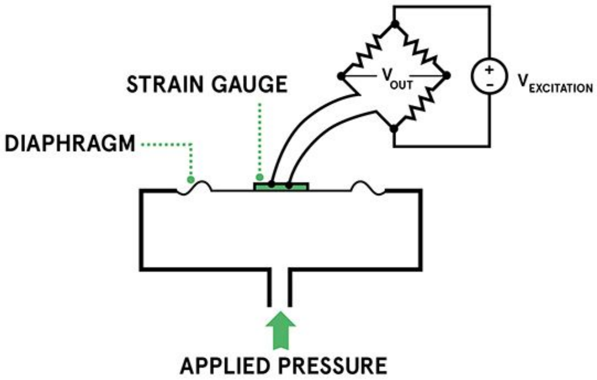


IMUs work, in part, by detecting changes in pitch, roll, and yaw.

Robotic Sensors

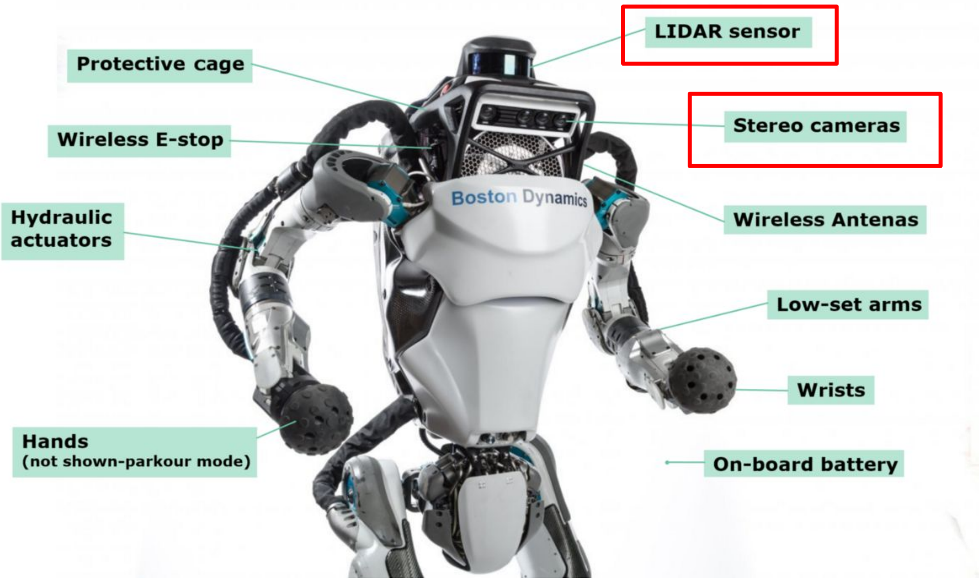
- Multimodal sensors are key to robot perception
    - Torque Sensor
        - Measure rotation force
    - Strain Gauge Pressure Sensor
        - Apply pressure and electronic signal changes
        - Touch sensor!





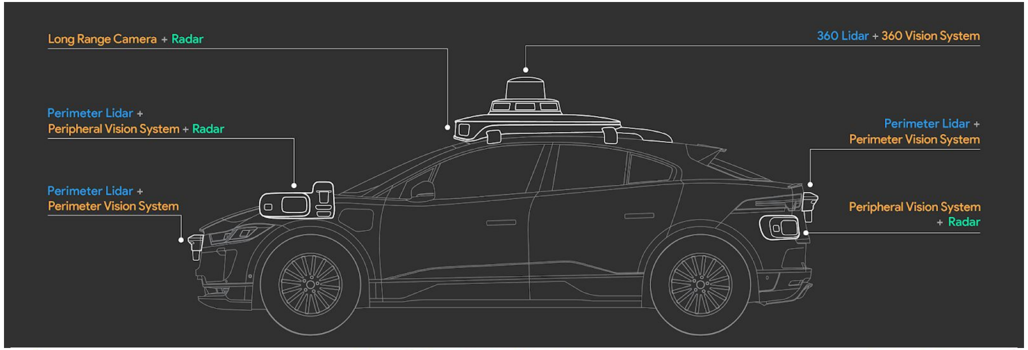
Robotic Sensors

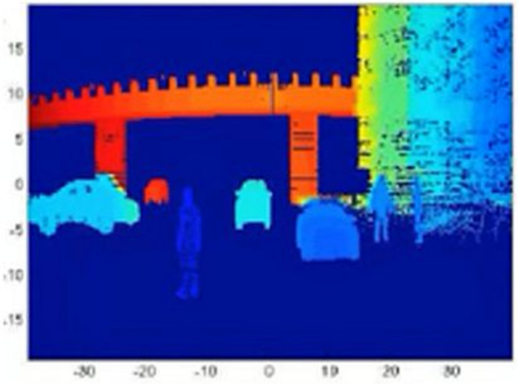
- Multimodal sensors are key to robot perception
  - Boston Dynamics Atlas
    - Cameras
    - Lidar
    - Inertial Measurement Unit (IMU)?



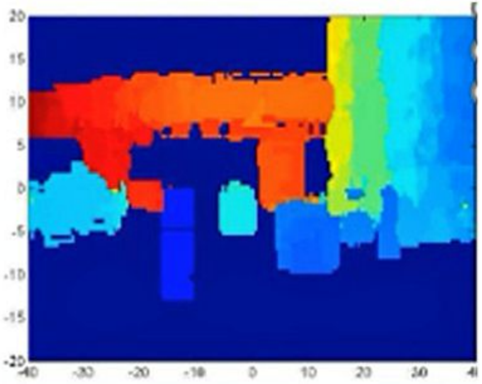
Robotic Sensors

- Multimodal sensors are key to robot perception
    - Waymo Self-driving Cars
        - Cameras
        - Lidar
            - Uses near-infrared
            - More precise
        - Radar
            - Uses microwave
            - Longer range





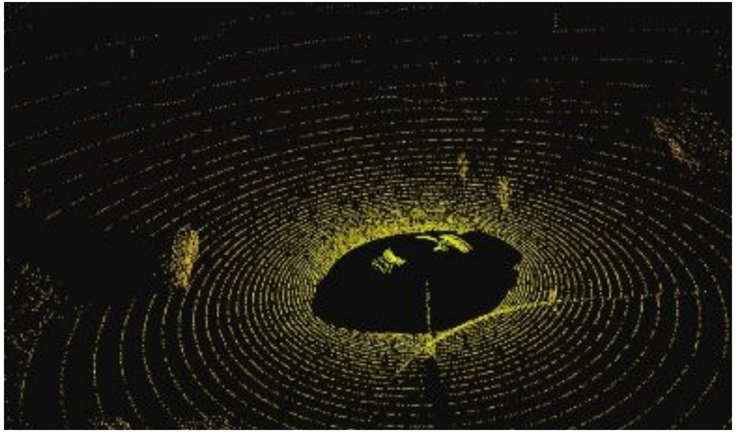
Lidar



High Resolution Radar

Robotic Sensors

- Multimodal sensors are key to robot perception
  - In 3D simulator CARLA
    - RGB Camera, IMU, Depth Camera, GPS, Radar, Lidar (you can set Lidar properties), Event Camera, etc.
      - Event camera: basically unlimited FPS (low-resolution) camera



*https://carla.readthedocs.io/en/latest/ref_sensors/

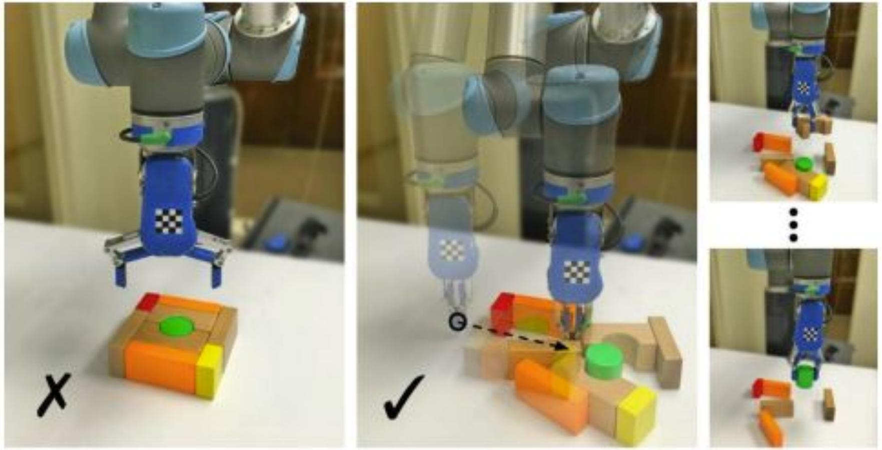
Agenda

- The Embodied AI paradigm
- Embodied System Design
- Sensors Overview
- Robot Vision vs. Computer Vision

Robot Vision vs. Computer Vision

- Robot vision is embodied, active and environmentally situated

[Detectron - Facebook AI Research]



[Zeng et al., IROS 2018]

Robot Vision vs. Computer Vision

- Robot vision is **embodied, active** and **environmentally situated**
- **Embodied**: Robots have physical bodies and experience the world directly. Their actions are part of a dynamical system together with the environment and have **immediate feedback** on their own sensation
- **Active**: Robots are **active perceivers**. They should know what and why it wishes to sense, and they should be able to choose what to perceive, and determine how, when and where to achieve that perception
- **Situated**: Robots are situated in the world. The environment will affect the whole system

Robot Vision vs. Computer Vision

- Robot vision is **embodied, active** and **environmentally situated**
  - Like human ego-vision

Ego4D dataset: https://ego4d-data.org/

Agenda

- The Embodied AI paradigm
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- Robot Vision vs. Computer Vision
- Recent robot demos from our lab

Our Lab: the Precognition Lab

香港科技大学(广州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

- Check our lab website for more cool demos: https://precognition.team/ Or check out our 小红书！

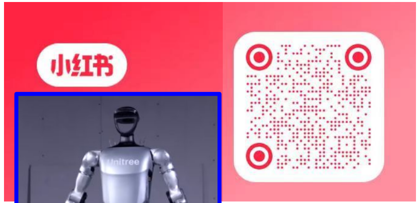
Navigation & Locomotion

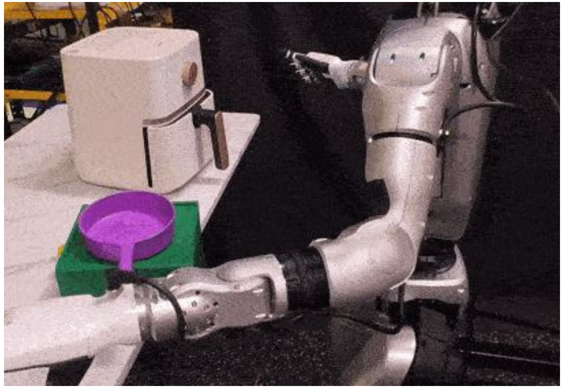
Machine Perception

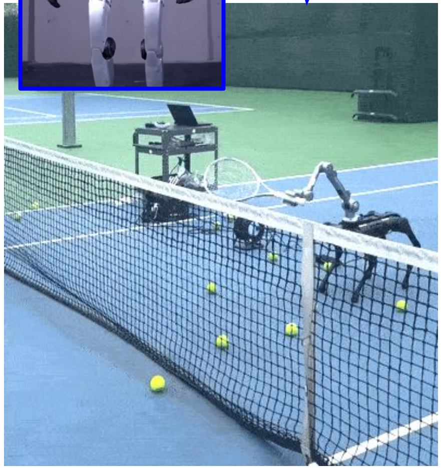
Embodied AI

Manipulation

Forecasting







Agile Locomotion

Manipulation with dex hand

Athletic Intelligence

Our robot demos: Social Navigation

- Check our lab website for more cool demos: https://precognition.team/ Or check out our 小红书！

Navigation & Locomotion

Machine Perception

Embodied AI

Manipulation

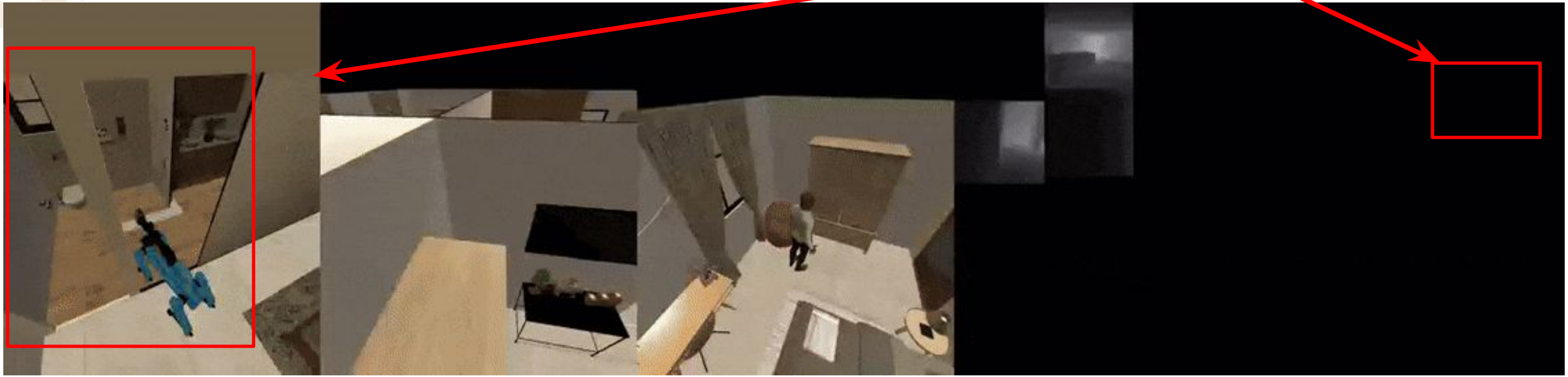
Forecasting

This tackles the problem of “how to make robot go somewhere you want”

Our robot demos: Social Navigation

Social Navigation using Human Trajectory Prediction [3]

- Predict human trajectory using visual information[1,2], make sure robot plans a path that follows social etiquette. Ensure physical and mental safety of AI robots
- **The robot is going to the plant.** See here how the robot moves backward to avoid the person

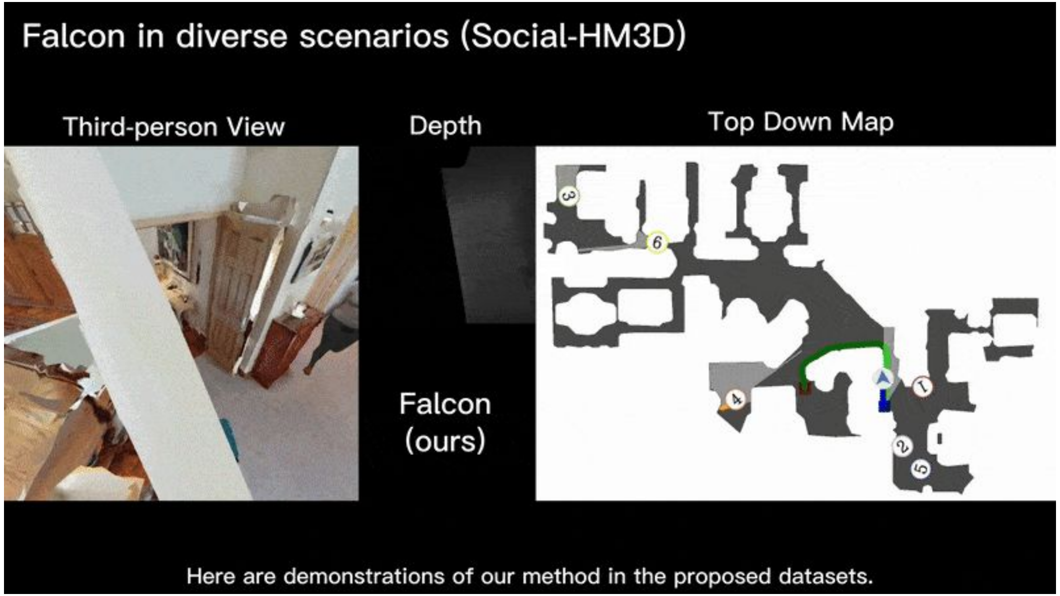


[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.
[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019
[3] “From Cognition to Precognition: A Future-Aware Framework for Social Navigation”, ICRA 2025

Our robot demos: Social Navigation

Social Navigation using Human Trajectory Prediction

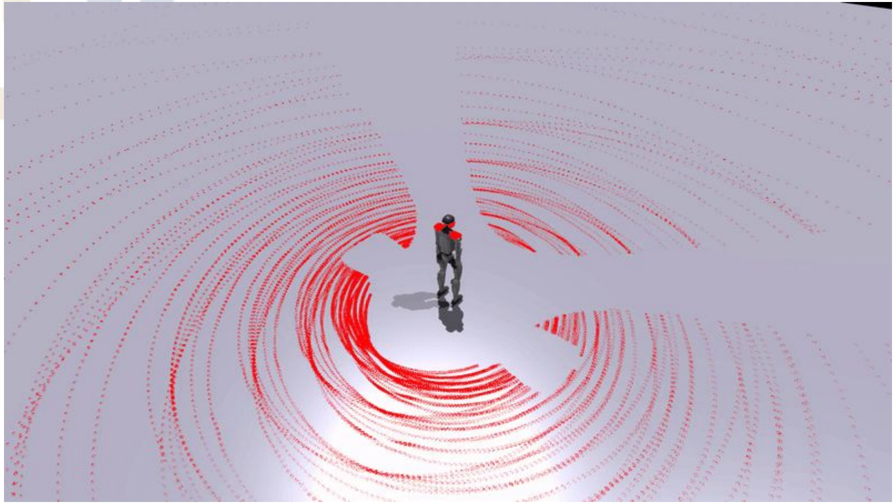
- More examples - You will be able to train this model yourselves for Project 3 homework!



[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.
[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019
[3] “From Cognition to Precognition: A Future-Aware Framework for Social Navigation”, ICRA 2025

Our robot demos: Agile Locomotion

- We can also utilize reinforcement learning with 3D simulator to train a “locomotion policy” that allows robot to avoid moving obstacles



*This is 0.5x playback speed

https://acodedog.github.io/OmniPerceptionPages/

Our robot demos: Manipulation

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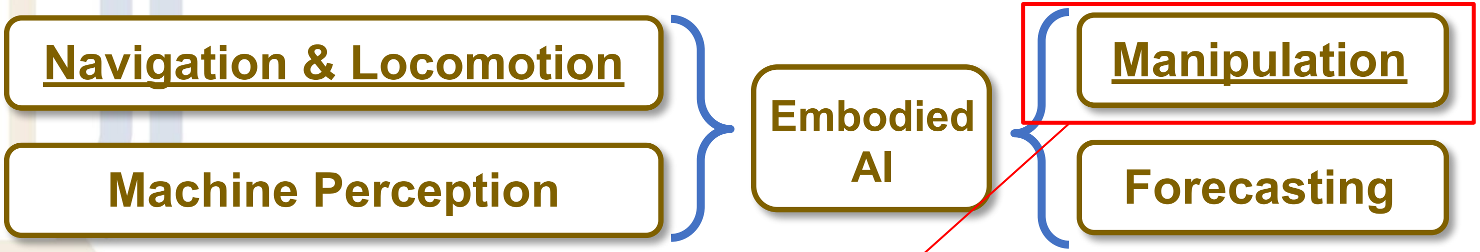
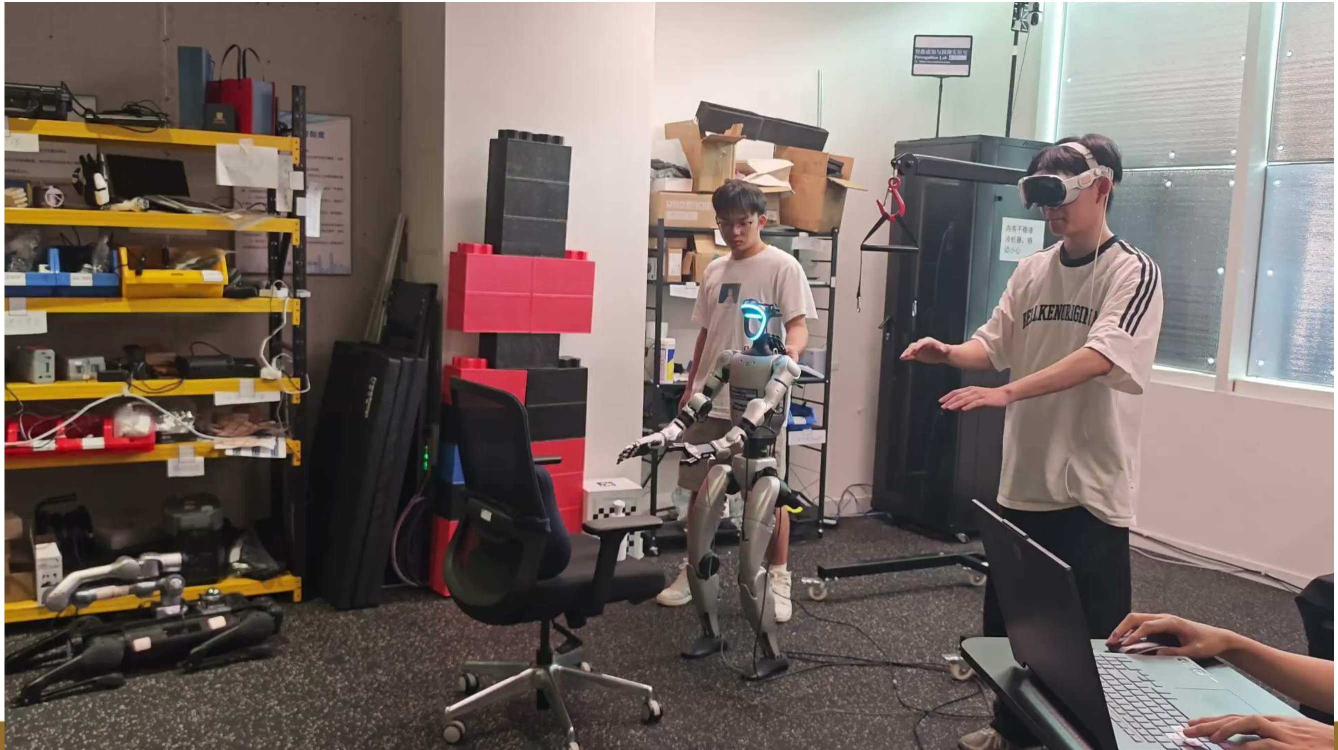


This tackles the problem of “how to make robot do something you want (with their hands)”

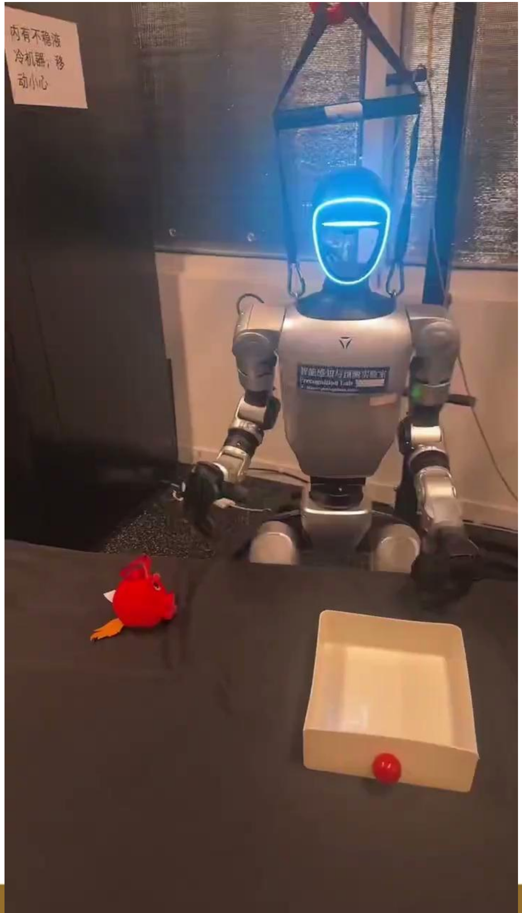
Our robot demos: Manipulation (Humanoid VLA)

- We can build a system that allows tele-operating of the whole-body of the humanoid robots, so we could collection training data



Our robot demos: Manipulation (Humanoid VLA)

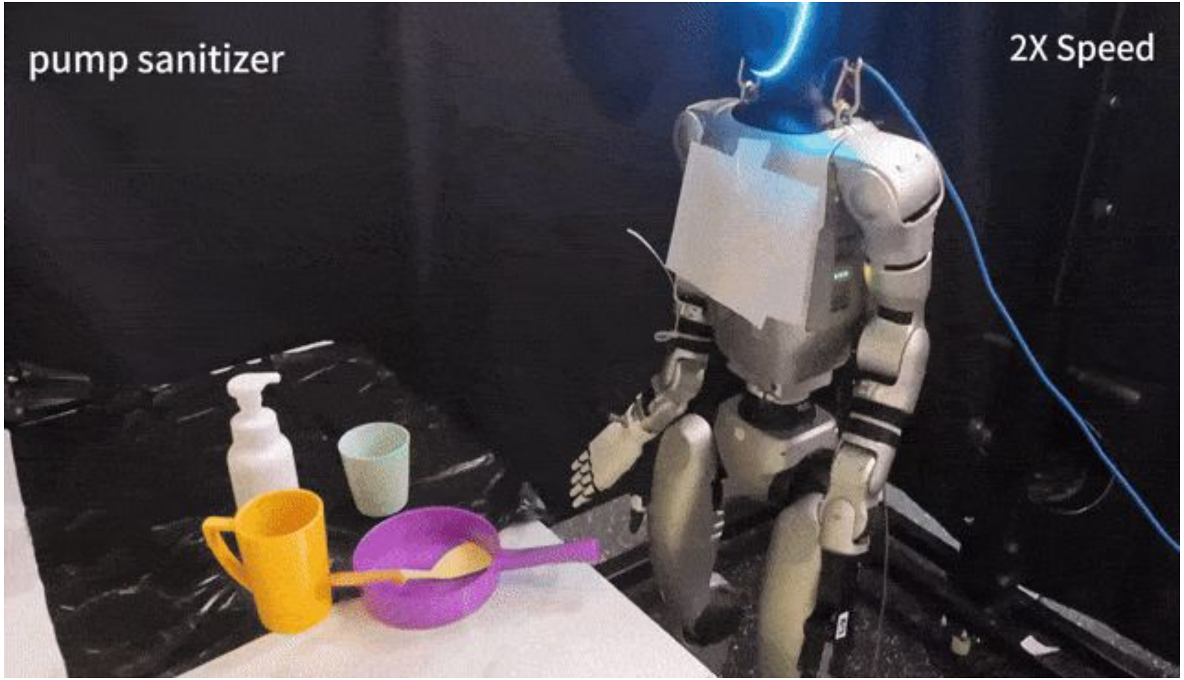
- So the robot can autonomously do tasks like “put the red bird into the box”
  - You can see that it moves very slow and may fail sometimes
  - Need a lot research work to improve!



Our robot demos: Manipulation (Humanoid VLA)

- We can even do zero-shot, open-vocabulary manipulation, with the help of vision-language large models

See how the robot handles the same object for different task instructions：
Pump sanitizer vs. pick up sanitizer



* GLOVER++: Unleashing the Potential of Affordance Learning from Human Behaviors for Robotic Manipulation. CoRL 2025

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- The Embodied AI paradigm
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- Sensors Overview
- Robot Vision vs. Computer Vision
- Robot Demos from my lab
- Next week - Machine Perception!

Reading Material

- Urmson, Chris, et al. "Tartan racing: A multi-modal approach to the darpa urban challenge." (2007).
- ROS tutorials: https://www.ros.org/blog/getting-started/

Thank you!